

An Exact Computer-Assisted Proof of the Dittert Conjecture for $n = 5$

Pedro Paulo Marques do Nascimento

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Review status. This document reports a new exact computational certificate produced in an interactive research session. The complete integer certificate and verifier are included. The argument has not yet undergone independent peer review and should be checked before being cited as an established result.

Abstract

For a nonnegative 5×5 matrix A whose entries sum to 5, let

$$\Phi(A) = \prod_{i=1}^5 r_i + \prod_{j=1}^5 c_j - \text{per}(A),$$

where r_i and c_j are the row and column sums. We give an exact computer-assisted proof that

$$\Phi(A) \leq 2 - \frac{5!}{5^5} = \frac{6130}{3125},$$

with equality only at the matrix J_5 whose entries are $1/5$. Published structural results reduce the boundary case to the face with two zero entries in distinct rows and columns. On that face an integer multiplier–sum-of-squares certificate proves a strict quintic inequality. The verifier checks all 80 730 quintic coefficients using arbitrary-precision integer arithmetic. The smallest residual coefficient is approximately $5.4206471 \cdot 10^{-5}$.

1 Statement and boundary reduction

Let

$$K_5 = \left\{ A = (a_{ij}) \in \mathbb{R}_{\geq 0}^{5 \times 5} : \sum_{i,j} a_{ij} = 5 \right\}.$$

Theorem 1. *For every $A \in K_5$,*

$$\Phi(A) \leq \frac{6130}{3125}.$$

Equality holds if and only if $A = J_5$.

We use two published structural facts. Hwang proved that the only possible positive Φ -maximizer is J_n [1]. Cheon and Wanless proved that a partly decomposable matrix cannot maximize Φ , and that the complete zero set of a maximizer cannot, up to row and column permutations, be one proper rectangular block [2, Theorems 3.2 and 3.7].

Lemma 2. *If a nonempty set of matrix positions contains no two positions in distinct rows and distinct columns, then all its positions lie in one row or all lie in one column.*

Proof. Choose (i, j) in the set. Every other position shares row i or column j with it. A position of each type would give two positions in distinct rows and columns, so only one type can occur. $\square$

Proposition 3. *It is enough to prove*

$$\Phi(A) < \frac{6130}{3125} \quad (A \in K_5, a_{11} = a_{22} = 0). \quad (1)$$

Proof. Compactness gives a global maximizer. A positive maximizer is J_5 by Hwang's theorem. If a boundary maximizer had no two independent zeroes, Lemma 2 would put its complete zero set in one row or column. If the whole row or column vanished the matrix would be partly decomposable; otherwise its zero set would be one proper rectangular block. Both cases are excluded by Cheon–Wanless. Row and column permutations therefore place two zeroes at $(1, 1)$ and $(2, 2)$, where (1) excludes the boundary maximizer. $\square$

2 The quintic on the two-zero face

Put $X = A/5$. Its entries sum to 1. On the face $x_{11} = x_{22} = 0$, index the remaining 23 entries by

$$V = \{(i, j) : 1 \leq i, j \leq 5\} \setminus \{(1, 1), (2, 2)\}.$$

Write $S = \sum_{v \in V} x_v$. Let $\mathcal{E}$ be the collection of five-element subsets of V that meet all five rows or all five columns, and put

$$F(x) = \sum_{E \in \mathcal{E}} \prod_{v \in E} x_v. \quad (2)$$

There are 2000 row transversals, 2000 column transversals, and 78 allowed permutation transversals, hence $|\mathcal{E}| = 3922$.

Expanding the row-sum and column-sum products shows that their common square-free monomials are exactly the permanent terms. Consequently

$$\prod_i R_i + \prod_j C_j - \text{per}(X) = F(x), \quad \Phi(A) = 5^5 F(x). \quad (3)$$

Thus (1) is equivalent to

$$G(x) := \frac{1226}{5^9} S^5 - F(x) > 0 \quad (x \geq 0, S = 1). \quad (4)$$

3 Exact certificate

Let $m_2(x)$ be the vector of all 276 quadratic monomials in the 23 variables. The automorphism group Γ of the two-zero face has order 144. Its action has three vertex orbits, of sizes 2, 12, and 9; choose representatives u_1, u_2, u_3 .

The certificate archive contains integer lower-triangular matrices

$$\hat{L}_p \in \mathbb{Z}^{276 \times 276} \quad (1 \leq p \leq 3),$$

and sets $L_p = 2^{-28} \widehat{L}_p$. The verifier expands the identity

$$G(x) = \sum_{p=1}^3 \sum_{\gamma \in \Gamma} x_{\gamma u_p} \left\| L_p^\top m_2(\gamma x) \right\|_2^2 + \frac{1}{D} \sum_{|\alpha|=5} R_\alpha x^\alpha, \quad (5)$$

$$D = 5^9 2^{56} = 140\,737\,488\,355\,328\,000\,000\,000.$$

It checks all

$$\binom{23+5-1}{5} = 80\,730$$

coefficients and finds

$$\min_{|\alpha|=5} R_\alpha = 7\,628\,882\,599\,067\,611\,080 > 0. \quad (6)$$

Proposition 4. *The strict inequality (4) holds on the nonnegative simplex $S = 1$.*

Proof. Every term in the first sum of (5) is nonnegative for $x \geq 0$: it is a nonnegative coordinate times a square norm. The residual polynomial has strictly positive coefficients by (5)–(6). Since $S = 1$, at least one coordinate is positive, and its fifth-power term makes the residual polynomial strictly positive. $\square$

Proposition 4 and Proposition 3 prove Theorem 1.

4 Exact verification and scope

The numerical semidefinite program was used only to discover the three Gram matrices. Each was shifted into the positive-definite cone, Cholesky factored, and rounded on a 2^{-28} grid. The saved integer factors are the proof object; no floating-point number is used by the verifier for a correctness decision.

The verifier independently reconstructs the face, the group, all 3922 hyperedges, the three Gram matrices, and every coefficient of (5). Run, without Python's `-O` option,

```
python3 verify_primary.py dittert_n5_exact_certificate.npz
```

The expected final line is **CERTIFIED**. The certificate SHA-256 digest is

373041bda29e1164059a2adbad1aaacd2911784a273629f419972e7ec7b5fdca.

The verifier establishes the finite polynomial identity exactly. It does not reprove the published structural theorems used in Proposition 3. This working proof has not yet undergone independent peer review.

References

- [1] S.-G. Hwang, *A note on a conjecture on permanents*, Linear Algebra and its Applications 76 (1986), 31–44.
- [2] G.-S. Cheon and I. M. Wanless, *Some results towards the Dittert conjecture on permanents*, Linear Algebra and its Applications 436 (2012), 791–801.